Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	6th November 2025	**Maximum Marks**	10+50
Course Code	CD252IA	**Duration**	120 Minutes
Sem	V	Faculty: PD /PT/AKB/NGD/SGD/CNS/SNM/VMN/KSK/Harshitha	

Database Management Systems (Common to CS, IS, CD, AI & CY)

Sl.No	PART-A	M	BT	CO
1.	In a University database the enrolment table records the courses taken by the students in various semesters. It includes attributes StudentID, CourseID and semester. Given that students may enroll in multiple courses across different semesters identify the attribute that serves as a partial key and justify the choice.	2	L3	2
2.	"Database supports multiple views of data". Give an example to justify the statement.	2	L2	2
3.	In a Hospital Management System, the Prescription entity records the details of medicines prescribed to patients during a consultation. Can this entity be considered a weak entity? Justify your answer.	2	L3	2
4.	Differentiate between Multivalued and composite attributes. Give an example representation for each.	2	L2	2
5.	Outline the importance of NULL value. Give two instances where a value of given attribute can be NULL.	2	L2	1

Sl. No.		PART-B	M	BT	CO
1		In a **Company Database**, consider the following schema: EMPLOYEE(Fname,Minit,Lname,Ssn,Bdate,Address,Sex,Salary,Super_ssn,Dno) DEPARTMENT(Dname,Dnumber,Mgr_ssn,Mgr_start_date) DEPT_LOCATIONS(Dnumber, Dlocation) PROJECT(Pname,Pnumber,Plocation,Dnum) WORKS_ON(Essn,Pno,Hours) DEPENDENT(Essn,Dependent_name,Sex,Bdate, Relationship) For each of the following update operations, analyze the potential constraint violations and suggest appropriate enforcement mechanisms or schema-level constraints that would prevent such violations. 1) What constraint violations may occur when deleting a project that is referenced in WORKS_ON? 2) What integrity checks are triggered when updating manager details in a department record? 3) What constraints must be satisfied when inserting a new employee record into a relational schema? 4) Discuss the referential integrity checks involved when assigning a manager to a department. Which constraints govern the update of work hours in a many-to-many relationship table like WORKS_ON?	10	L3	2
2	(a)	In a **University Database**, consider the following schema: DEPARTMENT(Dept_ID, Dept_Name, Location) STUDENT(Student_ID, Name, Dept_ID, CGPA) The Dept_ID in the STUDENT table is defined as a foreign key referencing the Dept_ID in the DEPARTMENT table. If an Update operation is performed UPDATE STUDENT SET Dept_ID = 'D999' WHERE Student_ID = 'S103'	06	L3	2

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		Identify and explain the type of constraint that will be violated if D999 does not exist in the DEPARTMENT table.			
	(b)	"The **Delete** operation can violate only referential integrity". Justify the statement with any example of your choice.	04	L2	2
3	(a)	With the help of a neat diagram, explain the three schema architecture.	06	L2	1
	(b)	Using the above architecture, outline the types of data independence.	04	L2	1
4		An educational institution wants to design an Online Course Management System to handle information related to students, courses, faculty, and enrolments. The system should store details about students and faculty members. Faculty members handles courses offered by a particular department. Each course is handled by one or more faculty members, and each faculty member can handle multiple courses. Students can enroll in multiple courses, and each course can have multiple students enrolled. For every enrolment, the system should record the Enrollment Date and Grade obtained by the student. The institution also wants to maintain information about departments, which have Department ID, Name, and Head of Department. Each department offers multiple courses and has multiple faculty members. Design an ER diagram for the above requirements. Capture all entities, attributes, primary keys, and relationships. Make suitable assumption wherever required and justify the same explicitly. Represent all participation types, cardinality ratios. Identify and represent different types of attributes. Justify your answer.	10	L4	2
5		For the above E-R Diagram, Apply the ER to Relational Mapping algorithm to convert ER to Relational Schema. Depict all steps appropriately and justify your answer.	10	L4	2

COURSE OUTCOMES:

CO1: Understand and explore the needs and concepts of relational, NoSQL database and Distributed Architecture

CO2: Apply the knowledge of logical database design principles to real time issues.

CO3: Analyse and design data base systems using relational, NoSQL and Big Data concepts

CO4: Develop applications using relational and NoSQL database

CO5: Demonstrate database applications using various technologies.

COs/BTL	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
Marks	12	38	--	--	--	--	20	20	20

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